

Does frustration change the dynamics of an elastic network model?

2026-08-25. Acylphosphatase, 2acy chain A: 98 residues, 960 contacts at $d_{\max} = 10.5$ Å, uniform $k_{ij} = 1$ (ANM). Everything seeded and reproducible from [analyses/](#).

1. The question

Every elastic network model in use fits its springs to a structure: $l_{ij} = d_{ij}^e$, so at the native conformation every spring sits exactly at its rest length. The model is *unfrustrated* by construction.

Real proteins are presumably not, and a mutant certainly is not: perturb the rest lengths, let the structure relax, and the springs no longer all reach their rest lengths at once — the network carries **strain**. Standard practice handles a mutant structure by refitting: take the coordinates, set $l_{ij} = d_{ij}^e$ again, diagonalise. That throws the strain away.

The assumption under test is that throwing it away does not matter for the motions. It is made constantly and, as far as I can tell, never checked — for the practical reason that an ordinary ENM cannot produce a genuinely strained state to check it on.

§2 builds a model that can. §§3–6 do the test.

2. A model whose parameters are the rest lengths

2.1 The potential

$$V(\mathbf{r}; \{l_{ij}\}) = V_0 + \frac{1}{2} \sum_{i < j} k_{ij}(l_{ij}) (d_{ij}(\mathbf{r}) - l_{ij})^2$$

The rest lengths are the only parameters. The single departure from a standard ENM is that the spring constant is a function of the **rest length**, $k_{ij}(l_{ij})$, not of the equilibrium distance $k_{ij}(d_{ij}^e)$.

At the wild type these are the same function, since the fit sets $l_{ij} = d_{ij}^e$. They differ only for a mutant, and then in a way that matters:

the contact map becomes a property of the parameters, not of the structure.

Contacts are made and broken by *mutation*, when an l_{ij} crosses the cutoff — not by relaxation. For the ANM, $k_{ij}(l_{ij}) = \mathbb{1}[l_{ij} \leq d_{\max}]$, with backbone neighbours

always bonded. A pair with $k_{ij} > 0$ is **active**; the active pairs are the contact map. $V_0 = 0$ throughout, so V of a state is exactly its strain energy.

2.2 The state

A state carries l_{ij} for **all** $N(N-1)/2 = 4753$ pairs, not just the 960 active ones, initialised to $l_{ij} = d_{ij}(\mathbf{r}_{\text{wt}}^e)$.

This is forced, not convenient. Every *active* wild-type pair has $l_{ij} \leq d_{\text{max}}$ by construction, so if only active pairs carried a rest length no inactive pair could ever be pulled inside the cutoff, and the network could only thin. Carrying all pairs makes breaking and forming symmetric. The cost is negligible: a force evaluation over 4753 pairs takes ~ 4 ms against 30 ms for one diagonalisation, and the Hessian stays $3N \times 3N$ because only active pairs contribute.

2.3 One mutation

Pick a site j ; perturb the rest length of every pair involving j by $\delta l_{ij} \sim \mathcal{N}(0, \sigma^2)$; recompute $k_{ij}(l_{ij})$; **minimise V numerically** for $\mathbf{r}_{\text{mut}}^e$; evaluate V exactly there. $\sigma = 0.3$ Å unless stated.

2.4 The discipline

No expansion ever conditions the dynamics. The structure comes from minimising V — not linear response, not $\mathbf{C}_{\text{wt}}\mathbf{f}$, not a two-term formula. Energies are read at the minimum. Normal modes are computed only to *observe* a state already accepted.

This is affordable: one exact minimisation converges in 4–20 Newton steps at ~ 0.25 s, so a trajectory runs at ~ 4 proposals per second. There is no performance argument for the linear-response shortcut.

2.5 Energy differences

The reference is always named, because along a trajectory it is a strained state, not the founder:

$$\begin{aligned}\Delta V_{\text{stress}} &= V_{\text{mut}}(\mathbf{r}_{\text{ref}}) - V_{\text{ref}}(\mathbf{r}_{\text{ref}}) \\ \Delta V_{\text{relax}} &= V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) - V_{\text{mut}}(\mathbf{r}_{\text{ref}}) \\ \Delta V_{\text{min}} &= V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) - V_{\text{ref}}(\mathbf{r}_{\text{ref}}^e) = \Delta V_{\text{stress}} + \Delta V_{\text{relax}}\end{aligned}$$

Two Hamiltonians at one structure; one Hamiltonian at two structures; the difference of minima, which is what a trajectory accepts on. All computed by evaluating Hamiltonians, never a closed form — at a strained reference a closed form needs a cross term, and here further terms because $k_{ij}^{\text{mut}} \neq k_{ij}^{\text{ref}}$.

Dropping the Δ is not a notational nicety. Fifteen substitutions along, $V_{\text{mut}}(\mathbf{r}_{\text{ref}}) = 7.90$ while $\Delta V_{\text{stress}} = 1.06$ — a sevenfold difference, exactly V_{ref} , growing with the walk.

2.6 Definitions

Strain, two measures: $s_{\text{max}} = \max_{ij} |d_{ij}(\mathbf{r}^e) - l_{ij}|$ in Å, and $E_{\text{strain}} = \frac{1}{2} \sum k_{ij} (d_{ij}(\mathbf{r}^e) - l_{ij})^2$, equal to V since $V_0 = 0$.

Frustration: no conformation relaxes every spring at once, i.e. $E_{\text{strain}} > 0$ at the minimum.

Hessian, for an active pair,

$$\mathbf{K}_{ij} = -k_{ij} [\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T + g_{ij} (\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T - \mathbf{I})], \quad g_{ij} = \frac{l_{ij}}{d_{ij}(\mathbf{r})} - 1$$

with $\mathbf{K}_{ii} = -\sum_{j \neq i} \mathbf{K}_{ij}$. The g_{ij} piece is the **transverse term**; it vanishes exactly when a spring is at its rest length, so **frustration reaches the dynamics only through g_{ij}** . Standard ENMs drop it because for them it is identically zero.

RMSF of site i : $\sqrt{\langle |\delta \mathbf{r}_i|^2 \rangle}$, from the 3×3 diagonal block of $\mathbf{C} = \beta^{-1} \mathbf{K}^+$.

RMSIP over $M = 20$ modes: $\sqrt{M^{-1} \sum_{a,b} (\mathbf{u}_a^A \cdot \mathbf{u}_b^B)^2}$. It scores a *subspace*.

Mode matching is a one-to-one assignment maximising total overlap (`clue::solve_LSAP`), not row-wise maxima, which are not a matching and inflate overlaps.

$\beta = 1$ wherever entropy appears — with $k_{ij} = 1$ this is a convention, not a physical temperature.

3. Generating strained states

Purpose. The test needs states whose springs are genuinely strained. This section produces them; it is method, not result.

What I did. From the wild type, repeatedly: draw a site uniformly, mutate it (§2.3), minimise exactly, accept with probability $\min\{1, e^{-\nu \Delta V_{\text{min}}}\}$ measured against the *current* state. A neutral walk sets $\nu = 0$ and accepts everything. 120 substitutions at three selection strengths, seeds $300 + 10\nu$.

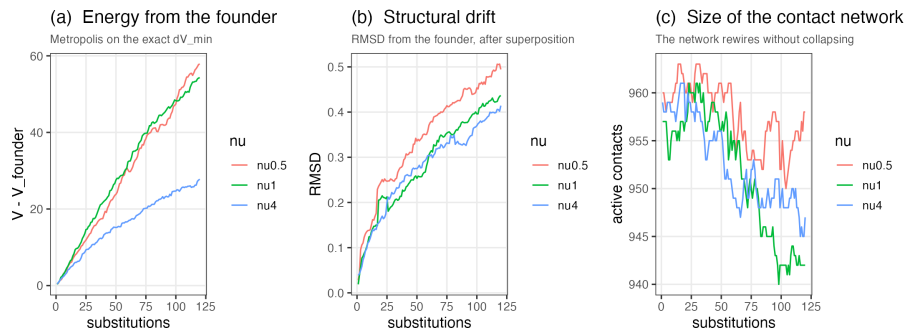


Figure 1. Three walks at $\nu = 0.5, 1, 4$ (colour), 120 accepted substitutions each, $\sigma = 0.3$. x is accepted substitutions, not proposals. **(a)** V relative to the founder — since $V_0 = 0$, the state’s strain energy. **(b)** RMSD from the founder after rigid-body superposition, Å. **(c)** Number of active pairs; the wild type has 960.

Result.

ν	final E_{strain}	final s_{max} (Å)	active pairs	acceptance
0.5	57.85	1.479	958	0.727
1	54.26	1.307	942	0.612
4	27.76	0.977	947	0.198

No run stalled. Selection sets the energy level; the network ends within 2 % of its starting size in every case.

What it means. The walks produce genuinely strained states — s_{max} of 1–1.5 Å against a 10.5 Å cutoff — and keep producing them. That is all this section is for. Two properties that had to hold for it to work at all (the contact map really moves; the walk does not run out of acceptable moves) are measured in Appendices A and B.

4. The test: rebuild a strained state and compare

Question. Take a strained state, refit an ENM to its coordinates the way everyone does, and ask how different the resulting dynamics is.

What I did. Three independent neutral walks of 60 substitutions (seeds 501–503), saving every 10 — **18 states**. At each, two models built **at the same coordinates**:

	rest lengths	active set	transverse term
frustrated	the state's own l_{ij}	from $k_{ij}(l_{ij})$	$g_{ij} \neq 0$
rebuilt	reset to d_{ij} here	re-derived from the structure	$g_{ij} = 0$

The rebuilt model uses only the coordinates: it is the standard construction. Both were verified at a true minimum with exactly six zero modes before anything was measured — off a stationary point the frustrated Hessian acquires spurious near-zero and negative eigenvalues, and $\text{RMSF} \sim 1/\lambda$ then diverges.

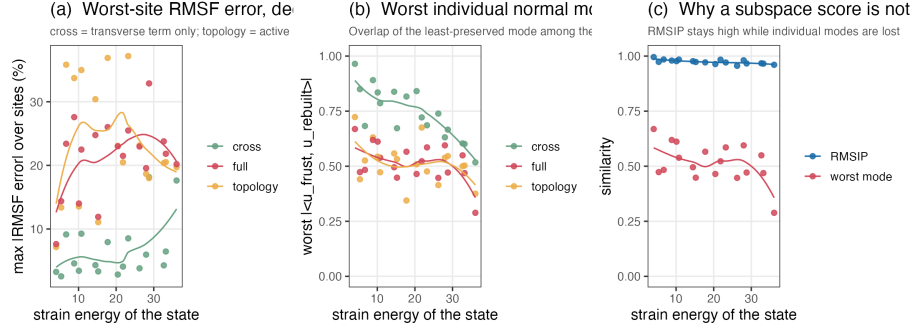


Figure 2. Rebuild error against accumulated strain, 18 states pooled from three walks (not 18 independent draws). x is E_{strain} . Colours decompose the difference: **green** = transverse term alone (same active set, $g_{ij} \neq 0$ vs $g_{ij} = 0$); **yellow** = active set alone (both $g_{ij} = 0$); **red** = both, the full rebuild. Lines are loess guides. **(a)** Largest relative RMSF error over the 98 sites. **(b)** Overlap of the least-preserved mode among the 20 softest. **(c)** That worst-mode overlap (red) beside RMSIP over all 20 (blue), full rebuild only.

Result.

	median over 18 states	range
RMSF profile correlation	0.982	0.961–0.995
worst-site RMSF error	22.7 %	7.6–32.9 %
RMSIP, 20 softest modes	0.973	0.956–0.995
block-5 mode overlap	0.928	0.844–0.965
max eigenvalue difference	20.1 %	up to 76.3 %
$TS_{\text{rebuilt}} - TS_{\text{frustrated}}$	−4.57	−6.34 to −0.44
net change in active pairs	+25	+1 to +30

Every one is monotone in substitution number ($|\text{cor}| \approx 0.87$), so these medians are properties of the sampling grid: sampled at 10–30 the median ΔTS is -2.09 , at 40–60 it is -5.53 . The trend against strain in figure 2 is what carries meaning.

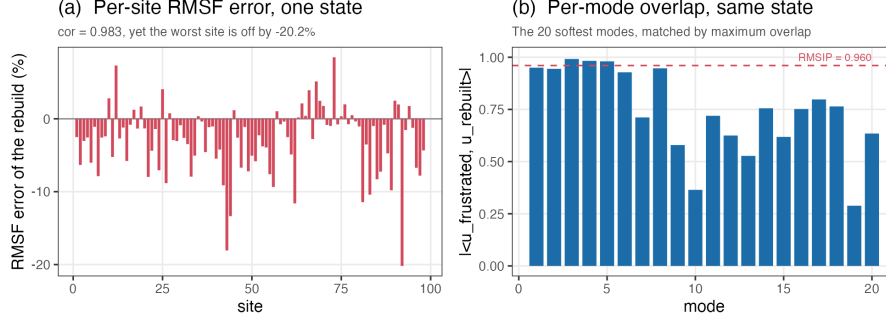


Figure 3. One state in detail — seed 501 at 60 substitutions, $E_{\text{strain}} = 36.1$, the *most strained* of the 18 and deliberately not typical. **(a)** Signed relative RMSF error at each of the 98 sites: the profile correlation is 0.983 and site 92 is still off by -20.2% . **(b)** Overlap of each of the 20 softest modes with its assigned partner; dashed line is RMSIP = 0.960. The panel shows what a subspace score averages over. It is **not** evidence that the rebuild destroys modes — see §6.

What it means, provisionally. Aggregate measures survive well; site-resolved ones are perturbed by tens of per cent; the entropy moves by several units. Two questions must be answered before any of that can be interpreted: *which* difference between the models causes it (§5), and whether an error of this size is even large (§6).

5. Is it the frustration, or the contact map?

Question. The rebuild changes two things at once: it zeroes the transverse term *and* re-derives the active set. Only the first is “neglecting frustration”. Which one does the damage?

What I did. Split the difference into the two steps and measure each. On the Hessian the split is exact by construction,

$$(\mathbf{K}_{\text{frust}} - \mathbf{K}_{g=0}) + (\mathbf{K}_{g=0} - \mathbf{K}_{\text{rebuild}}) = \mathbf{K}_{\text{frust}} - \mathbf{K}_{\text{rebuild}}$$

verified to 2.2×10^{-16} , so Frobenius norms of the two pieces compare fairly. Then — suspecting the answer might depend on the ANM’s hard cutoff — I repeated the whole decomposition with $k_{ij}(l_{ij})$ replaced by a sigmoid of width w ,

$$k_{ij}(l_{ij}) = [1 + \exp((l_{ij} - d_{\max})/w)]^{-1}$$

applied to **both** models, so they still differ only in l_{ij} and what follows from it.

Result. Under the ANM step, the active-set change is much the larger:

	transverse	active set	full	$\sqrt{\text{tr.}^2 + \text{a.s.}^2}$
median	2.53	15.09	15.31	15.38

The two add in quadrature to within 0.34 %, so they are nearly orthogonal perturbations, and the ratio is 5.6.

But **every edge on which the two active sets disagree is a marginal contact**. At seed 501/40, of 59 differing edges: gained pairs lie inside the cutoff by a median of 0.21 Å, lost pairs outside it by 0.09 Å, **100 % lie within 1.0 Å of $d_{\max}$** , 90 % within 0.5 Å, and together they hold 5.9 % of the state’s energy. The step function counts each at full strength $k_{ij} = 1$, so a pair at 10.4999 Å enters the Hessian as heavily as a core contact.

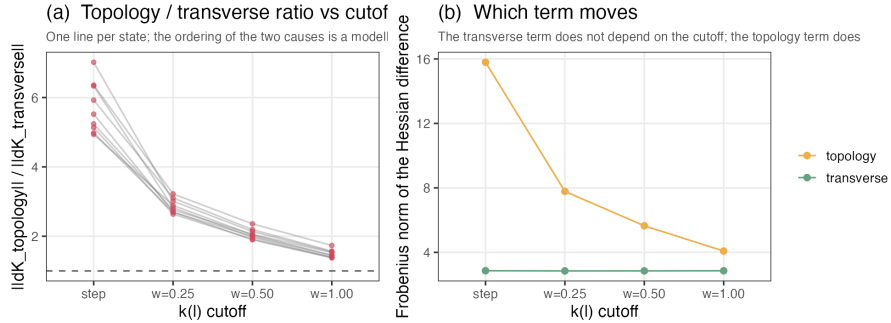


Figure 4. The same three walks sampled every 20 (9 states), re-expressed under four kernels. (a) One grey line per state across the four cutoffs; y is $\|\Delta \mathbf{K}_{\text{active set}}\|_F / \|\Delta \mathbf{K}_{\text{transverse}}\|_F$, dashed line at 1. (b) The two norms separately: transverse (green) flat by construction, active set (yellow) falling four-fold.

cutoff	transverse	active set	ratio
step	2.86	15.80	5.52
sigmoid, $w = 0.25$	2.85	7.78	2.82
sigmoid, $w = 0.50$	2.85	5.65	2.04
sigmoid, $w = 1.00$	2.86	4.08	1.45

Monotone in w in all 9 states.

What it means. *Under a hard cutoff*, re-deriving the contact map perturbs the Hessian about $5.5\times$ more than neglecting frustration does. But that factor is a property of the ANM step function, not of the protein: soften the kernel and the two become comparable. The transverse term is untouched throughout, as it must be, since it does not involve the cutoff.

The honest statement is that the two effects are of comparable size, and any claim that one dominates is a claim about the kernel.

6. Is the error large? A null control

Question. “The worst site is off by 23 %” means nothing without a scale. Is that a lot, for a perturbation of this size, in a spectrum this dense?

What I did. Built a perturbation matched in magnitude and null in kind: multiply the state’s own spring constants by $1 + sz$, with z a fixed vector of standard normals and s tuned by bisection until $\max |\Delta \mathbf{K}|$ matches the rebuild’s to within 5 %. It changes no rest length, adds no frustration, and leaves the active set intact (verified for all 27 draws). Three null draws per state, 9 states.

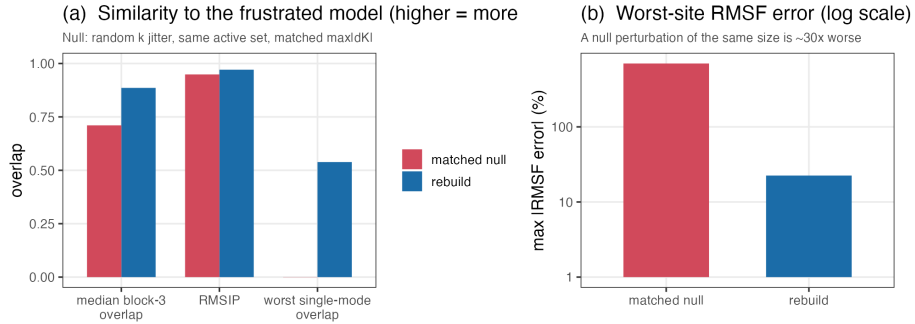


Figure 5. Rebuild (blue) against matched null (red). **(a)** Similarity to the frustrated model, higher = more similar: worst single-mode overlap, median block-3 overlap, RMSIP. **(b)** Worst-site RMSF error, log axis.

Result.

	rebuild	matched null
worst single-mode overlap	0.539	0.001
RMSIP over 20 modes	0.971	0.949
median block-3 overlap	0.886	0.710
worst-site RMSF error	22.5 %	698 %
max eigenvalue difference	0.221	0.997

What it means. A perturbation of the same size but arbitrary direction is some thirty times more destructive. **The rebuild is a remarkably gentle perturbation for its magnitude** — not merely “not too bad”, but far better than chance.

This also disposes of the individual mode overlaps. The rebuild shifts eigenvalues by ~15 % while adjacent levels are separated by ~6 %; a perturbation larger than the level spacing mixes eigenvectors within near-degenerate groups, and which vector is called “mode 10” becomes bookkeeping. The null confirms it: *any* perturbation of this size scrambles individual modes, and worse. Nothing mode-resolved in §4 is evidence about frustration.

7. Conclusion

Frustration can be neglected, and standard practice is better justified than this exploration set out to test.

Refitting an ENM to a strained structure preserves the RMSF profile (correlation 0.98), preserves the soft-mode subspace (RMSIP 0.97, block-5 overlap 0.93), and does so far better than a random perturbation of the same magnitude would (§6). What it costs is site-resolved accuracy — tens of per cent at the worst sites — and a few units of TS .

Two qualifications matter more than the headline:

1. **Most of the difference is not frustration at all**, but the contact map being re-derived. Under a hard cutoff that term is $5.5\times$ the frustration term; under a soft one they are comparable (§5). Which dominates is a modelling choice, usually made without comment.
2. **Individual normal modes of a perturbed ENM should not be read** — in this or any model with a spectrum this dense. The null control shows the low overlaps carry no information (§6).

The one place the contact map clearly does matter is entropy: letting k_{ij} follow l_{ij} widens the range of $\Delta(TS)$ across sites 6.25-fold while leaving the ΔV profile essentially unchanged (correlation 0.974, Appendix C). Adding or removing a spring changes the spectrum far more than moving a rest length does.

What would settle it. One protein, one cutoff, one k model. The obvious next step is §5’s decomposition under a smooth kernel throughout, where the frustration term may well become the interesting one.

Appendix A. The contact map moves, routinely

Needed for §3: if $k_{ij}(l_{ij})$ never changed the active set, the model would be an ordinary ENM with extra bookkeeping.

What I did. From the wild type, mutate one uniformly chosen site (§2.3), recompute k_{ij} , compare active sets. No minimisation — this is a property of the parameters alone. 200 draws at each σ , seed 1.

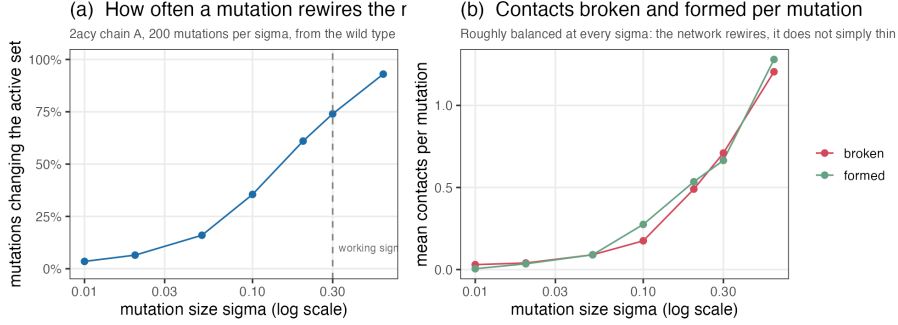


Figure A1. (a) Fraction of the 200 draws changing the active set, log x ; dashed line at the working $\sigma = 0.3$. (b) Mean pairs gained (green) and lost (red) per mutation.

σ	changing the active set	broken	formed
0.01	3.5 %	0.030	0.005
0.10	35.5 %	0.175	0.275
0.30	74 %	0.71	0.67
0.60	93 %	1.21	1.28

At the working σ , three-quarters of mutations rewire the network, and gains and losses are balanced — it rewires rather than erodes.

Appendix B. Strain opens a downhill channel

Needed for §3: if every mutation were uphill the walk would stall and there would be no strained states to test.

An earlier exploration (`sclfenm/MODEL.md` §6) proves that from a *relaxed* network, $\Delta V_{\min} \geq 0$ always. Its hypothesis fails here after the first mutation, so the question is empirical.

What I did. One neutral walk, 40 substitutions, seed 2, saving every 10. From each saved state (and from the wild type), 200 fresh trial mutations, each minimised exactly, $\Delta V_{\min}$ recorded and the trial discarded.

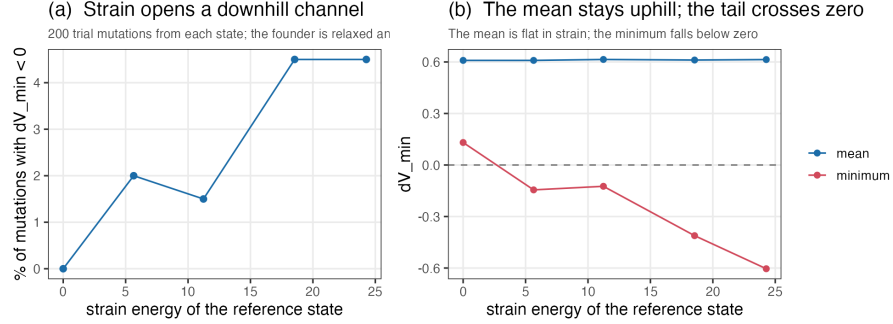


Figure B1. x is E_{strain} of the reference state. **(a)** Percent of the 200 trials with $\Delta V_{\text{min}} < 0$. **(b)** Mean (blue) and minimum (red) of the same 200 values.

substitutions	E_{strain}	mean ΔV_{min}	min	fraction < 0
0 (founder)	0.00	+0.609	+0.131	0.000
20	11.23	+0.615	-0.124	0.015
40	24.28	+0.614	-0.604	0.045

From the relaxed founder not one of 200 mutations is downhill; from a strained state a few per cent are. The mean is unmoved at ~ 0.61 throughout: **strain widens the distribution rather than shifting it**, so the walk keeps finding acceptable moves without drifting downhill.

Appendix C. What $k(l)$ does to a single-mutation scan

What I did. Mutate each of the 98 sites 8 times from the fixed wild type ($\sigma = 0.3$, seed 11), minimise each exactly, discard, average per site. Repeat with k_{ij} frozen at wild-type values — same seed, hence the same perturbations — so the two differ only through $k(l)$.

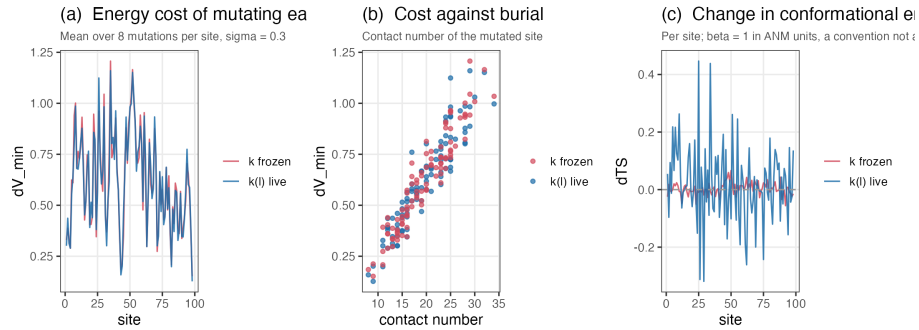


Figure C1. Blue = $k_{ij}(l_{ij})$ live, red = frozen. **(a)** ΔV_{min} per site. **(b)** The same against the mutated site's contact number. **(c)** $\Delta(TS)$ per site, $\beta = 1$.

(over the 98 sites)	$k_{ij}(l_{ij})$ live	k_{ij} frozen
$\text{cor}(\Delta V_{\min}, cn_j)$	+0.928	+0.946
$\text{cor}(\delta \mathbf{r}_j ^2, cn_j)$	-0.657	-0.712
range of $\Delta(TS)$	0.765	0.122

Both reproduce the standard results — buried sites cost more to mutate and move less. The energy profiles agree closely (correlation 0.974). The entropy range differs 6.25-fold: this is where the contact map shows up.

Appendix D. A smooth kernel

The step makes V discontinuous in the parameters. Replacing it with the sigmoid of §5 and running 40 substitutions at $\nu = 1$:

	E_{strain}	s_{max} over $k > 0.5$ (Å)	acceptance
$w = 0.25$	20.97	0.810	0.571
$w = 0.50$	21.49	1.057	0.482

Not comparable with §3’s table: these ran 40 substitutions against 120, and the sigmoid gives *every* pair $k_{ij} > 0$, so the network is fully connected (4753 pairs against 960). A proper comparison needs the sigmoid thresholded so the densities match. The kernel’s effect where it does change a conclusion is reported in §5.

Appendix E. Verification

Three suites in `checks/`, all passing.

test_core.R — the wild-type state reproduces penm’s ANM exactly (960 edges, $\max |\Delta \mathbf{K}| = 3.6 \times 10^{-15}$ on a scale of 12.9); the minimiser reaches $|\mathbf{F}| < 10^{-8}$ with exactly 6 zero modes; ΔV splits exactly at both relaxed and strained references; $k_{ij} = k_{ij}(l_{ij})$ holds after mutation; contacts break *and* form; mutation is exactly reversible in l , k , internal structure and V .

test_profiles.R — the frustrated Hessian genuinely differs from the $g = 0$ one; six zero modes wherever a spectrum is taken; the rebuild is relaxed by construction and does change the active set; mode matching is one-to-one; every accepted trajectory state is at its own minimum.

test_can_fail.R — six deliberate sabotages, all detected. Including a numerical-Hessian arbitration of the transverse sign on a *strained* network (correct form off by 3.65×10^{-6} , flipped by 0.245), and a demonstration that off a stationary point the spectrum shows 4 zero modes and a negative eigenvalue of -4.8×10^{-4} .

Appendix F. What I got wrong

Recorded because the corrections changed conclusions, not only numbers.

The minimiser returned unconverged structures. One draw in 200 from a strained state hit the iteration limit at $|\mathbf{F}| = 2.4 \times 10^3$ and yielded $\Delta V = 1.6 \times 10^5$, poisoning a mean into 778 where every other value was 0.61. It now raises an error.

ΔV_{stress} used the wild type’s k_{ij} — the LFENM assumption $\delta k = 0$, false by construction here. It made a 26 % discrepancy look like truncation error; with k_{ij}^{mut} the gap is 0.009 %. I then “fixed” the failing check by freezing k , bending the experiment to suit an expansion this study exists to avoid.

I concluded “topology dominates frustration by 5.6×” without testing whether the ratio depended on the kernel. It does, completely (§5).

I read low mode overlaps as physical. They are not; a null control shows any perturbation of that size does worse (§6).

Mode matching used row-wise maxima, not a matching: 31 of 54 comparisons had two modes claiming one partner, inflating every overlap.

I quoted a block-5 overlap of 0.84 as “the defensible claim”. It is the worst of 18 states; the median is 0.93.

Smaller: a claim that all wild-type $l_{ij} \leq d_{\text{max}}$ (true only of active pairs — over all 4753 they reach 36.3 Å); a mechanism asserting active pairs “cannot drift” (falsified by 66 lost edges); an unsourced count “318 to 228” that no script produces; a transcription error in the sabotage figures.

Appendix G. Limitations

- One protein, one cutoff, one k model, one mutation size for the main result.
- $\beta = 1$ in ANM units is a convention; absolute TS values are not physical.
- The 18 states are 3 walks $\times$ 6 correlated snapshots, not 18 independent draws; effective n for trajectory-level statements is 3.
- Mutation perturbs all of a site’s pairs (**radius = Inf**), so one mutation can reach across the protein.
- Trajectories are 40–120 substitutions at a single temperature; no phylogeny.
- The penm package is not modified. The frustrated Hessian is reimplemented here because **set_enm()** blocks **frustrated = TRUE**.

Appendix H. Files

R/klfenm_core.R	state, $k(l)$, energy, Hessian, exact minimiser, mutation
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R/klfenm_profiles.R	per-site and per-mode comparisons
R/klfenm_trajectory.R	scans and walks
analyses/run_all.R	§§3–5 and Appendices A–D → data/results.rds (~45 min)
analyses/run_control.R	§6 → data/control.rds
figures/make_figures.R	the eight figures
checks/	test_core.R, test_profiles.R, test_can_fail.R

Rscript analyses/run_all.R
Rscript analyses/run_control.R
Rscript figures/make_figures.R